

Objective function

Linearity and nonlinearity

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Optimization: principles and algorithms



ÉCOLE POLYTECHNIQUE
FÉDÉRALE DE LAUSANNE

Linearity

*A great deal of simplification comes when the function is linear.
When it is not, the difficulty depends on the type of non linearity.*

Linear functions

$$f : \mathbb{R}^n \rightarrow \mathbb{R} : f(x) = \sum_{i=1}^n c_i x_i = c^T x, \quad c \in \mathbb{R}^n$$

$$f : \mathbb{R}^n \rightarrow \mathbb{R}^m : f(x) = Ax, \quad A \in \mathbb{R}^{m \times n}$$

Affine functions

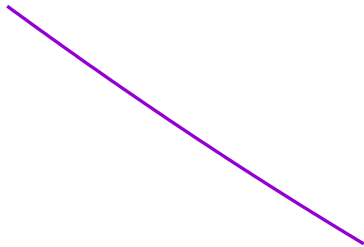
$$f(x) = c^T x + d, \quad d \in \mathbb{R}$$

$$f(x) = Ax + b, \quad b \in \mathbb{R}^m$$

In optimization, the constant do not matter, and “linear” may be used for “affine”

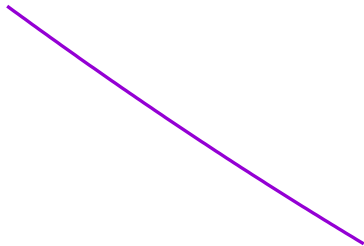
Nonlinear functions

$$f(x) = \frac{x^2}{100} - 2x + 1$$

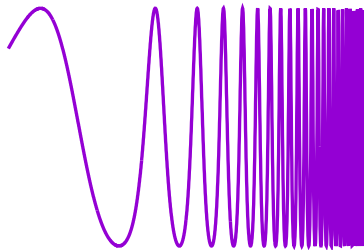


Nonlinear functions

$$f(x) = \frac{x^2}{100} - 2x + 1$$



$$f(x) = \exp(\cos(e^{-x}))$$



Lipschitz continuity of the gradient

$\nabla f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ is Lipschitz continuous if there exists a $M > 0$ such that, for all x, y :

$$\|\nabla f(x) - \nabla f(y)\| \leq M\|x - y\|$$

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M : Lipschitz constant

Summary

We will treat linear and nonlinear optimization separately.